
Forecasting Is Not World Modeling: Accuracy, Action Fidelity, and an Evaluation Matrix for a Task-Critical Thermal Process

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Abstract

Task-critical industrial processes demand more from a temporal model than accurate forecasts. A model promoted to the inner loop of a controller must respond to candidate actions with the right sign, magnitude, and transient shape, and it must know when it cannot. We study this gap on a supercritical once-through boiler, using 82 days of dual-side field data. A deep forecasting model attains 0.7–0.9 °C MAE at 18 steps, yet its valve-step responses span only 0.00004–0.040 °C with unstable direction across folds, valves, and seeds—functionally disabling the action channel. A physics-structured world model with the same data trades accuracy (about 3.5 °C) for an action channel that responds at one step but inverts sign at 60 steps—an inversion we localize to one learned feedback term (zeroing it restores the correct direction on 100% of windows). We report both findings through a discrimination matrix: an identifiability gate on same-type synthetic data, per-layer structural ablations with budget and convergence diagnostics, and an action-blindness and leakage audit. Finally we locate the physics model’s accuracy cost: after proper training it approaches persistence at the terminal stage (0.16 vs. 0.09 °C one-step MAE, beating persistence at 18 steps) while the upstream stage fails 38× worse than persistence, and the earlier monotone regime bias collapses to ± 0.2 °C—a training artefact, not a parameter-count one. Our results support verifiability, not accuracy, as the qualifying property of a world model, and a two-tier architecture in which a learned proposer is bounded by a certifiable reduced-order model. Matrix verdicts are single-side (side A); the side-B matrix is deferred to a later stage.

1 Introduction

A temporal model that forecasts well is not automatically a world model. World models must maintain a state, imagine futures under interventions, and support decisions [8, 9, 5]. On safety-critical plants this qualification question precedes accuracy: a facility can tolerate a conservative heuristic because it is bounded; it cannot tolerate a model that predicts confidently outside its own validity [7]. Closed-loop identification warned decades ago that passively learned dynamics absorb the confounded structure of the controller that generated the data [6]. Recent evidence that time-series foundation models collapse under feedback [14], despite strong open-loop accuracy [1, 4], suggests the failure is general, not anecdotal.

We examine the question on a supercritical once-through boiler, a task-critical thermal process: superheated steam temperature must stay inside a narrow band while disturbances (load ramps, coal variation, soot blowing) and two spray attempermentation valves act on it. We train two lines on the same dual-side field data. The first is a deep forecaster with multi-step heads; it reaches 0.7–0.9 °C MAE

at 18 steps. The second is a physics-structured world model (a lumped-enthalpy state-space with thermodynamic property tables and learnable parametric closures). The forecaster wins on accuracy; the world model wins on what the forecaster never learned: responding to actions. This tension is not a property of our two implementations. It follows from what the two objectives optimize, and we measure it with a protocol designed to be reused: the discrimination matrix.

Our contributions are (i) an empirical demonstration, on dual-side field data, that forecast accuracy does not transfer to world-model qualification; (ii) an evaluation matrix for such qualification (identifiability gate, per-layer ablations, action-blind and leakage audits); and (iii) a diagnosis of where the physics model’s accuracy cost comes from, with quantified floors and a convergent agenda.

2 Testbed and Two Model Lines

2.1 The process

The unit (660 MW class, supercritical, once-through) drives steam through a chain of superheater stages interleaved with two spray attenuators: low-temperature superheater, first spray, platen superheater, second spray, final superheater. Five temperatures along this chain are observed at 10 s resolution, together with boundary channels (steam flow, coal command, separator pressure and temperature, feedwater temperature, outlet pressure, spray flow) and two valve positions. The valves are cross-paired: valve A governs side-B temperatures and vice versa. Data span 82 days per side; the windows used here are entirely supercritical dry operation. Fast temperature increments are small (0.09–0.45 °C per step), so the process is strongly autoregressive; load-driven swings are orders of magnitude larger.

2.2 Line 1: a deep forecaster

Line 1 is a direct multi-step forecaster (sequence model over a history window with dedicated heads per horizon step, trained with a Gaussian likelihood). It is representative of the pure forecasting line rather than a state of the art: the horizon heads are teacher-forced, not recursive. We state this plainly because the comparison that follows is about what the forecasting objective rewards, and the literature suggests recursive foundation forecasters inherit the same closed-loop failure [14]. Its 18-step MAE on held-out windows is 0.712 (side A) and 0.927 (side B).

2.3 Line 2: a physics-structured world model

Line 2 embeds the process structure: three lumped-enthalpy segments whose dynamics obey energy conservation with IAPWS-97 property tables, two spray injections with transport delays, metal-temperature coupling, and learnable parameters (segment conductances, evaporation gains, spray efficiencies), in the universal-differential-equation tradition of embedding structure while learning parameters [3, 11]. An action-blind closure absorbs unmeasured disturbances from state and whitelisted boundary channels only. A probabilistic observer estimates the initial state. The model rolls out recursively, so any counterfactual branch is a consistent trajectory of the same dynamics. Its 18-step MAE is about 3.5 °C (v0.1 protocol); retrained values under the v0.2 budget are reported in Section 4.

3 The Tension, Measured Three Ways

3.1 Action response: one line is functionally blind

We probe both models with $\pm 2\%$ valve steps, holding all else fixed, and compare against a data-anchored mixing reference (spray-flow regressions on the field data, $dW/dv = 27.8$ and 70.0 t/h per full travel for the two valves, combined with an energy balance at the spray station; the reference spans 0.21–0.59 °C for the primary valve and 0.53–1.48 °C for the secondary valve per +2%, depending on the steam-flow band). Line 1 responds with mean magnitudes of 0.00004–0.040 °C across folds, valves, and directions, with the correct sign in only one of eight fold–valve–direction cells at 3/3 consistency—two to four orders of magnitude below the reference, and functionally blind (Fig. 3a). Line 2’s one-step response lands on the reference scale. Structural honesty requires the rest of the

83 story. The response is partly a construction guarantee: the action enters an instantaneous mixing
84 equation that the output map re-evaluates at every step, so the model must respond at step one whether
85 or not the learned dynamics agree. The real plant’s one-step response is effectively zero ($+0.038$
86 and -0.041 °C for secondary-valve open/close events, $n = 22/48$) and builds over minutes; Line
87 2 therefore overshoots the plant’s same-scale response by an order of magnitude ($10\text{--}30\times$) at step
88 one. And at 60 steps the direction breaks: only 19–34% of windows end negative (the direction rule
89 as frozen required 100%), with a mean terminal delta of $+0.20$ to $+0.26$ °C. An ablation localizes
90 the inversion to one named term: the wall-rewetting feedback. More spray raises the droplet mass,
91 which lowers the dry-out factor, which raises the rewetting heat flux into the steam; the learned
92 rewetting gain over-compensates the direct cooling, and terminal temperature rises. Zeroing the
93 rewetting coefficient restores the correct sign on 100% of windows (mean -0.44 °C). The closure
94 is not responsible (identical probes with and without it). Line 2’s action channel therefore exists
95 by structure but is not yet faithful by physics: a controller using Line 1 as its internal model sees
96 a plant that does not react to its valves; one using Line 2 sees a plant that reacts immediately and
97 wrongly at long horizons. Both lines fail the same requirement—no mechanism certifies the action’s
98 sign—through different paths.

99 3.2 Counterfactual coherence

100 Beyond steady responses, an internal model must carry physically coherent counterfactual branches.
101 Line 1’s teacher-forced heads cannot propagate a perturbation through time; Line 2’s recursive state
102 does so by construction, and its boundary dependence is whitelisted so that no action leaks into the
103 disturbance path (Section 4, R1). The confounding is measurable in principle with contemporaneous
104 causal discovery on the closed-loop data [12]; our probes below are the model-side equivalent.

105 3.3 Accuracy: the cost is real, located, and mostly structural

106 Line 2 pays a $3\text{--}4\times$ aggregate accuracy price (H18 MAE 3.9 °C vs. $0.7\text{--}0.9$ for Line 1), but the price
107 is not uniform across the steam path. At the terminal stage the retrained model is nearly as good as
108 persistence one step ahead (0.16 vs. 0.09 °C MAE) and beats it at 18 steps (1.05 vs. 1.18); at the
109 upstream stage it is $38\times$ worse than persistence (9.4 vs. 0.25 °C), and the model’s learned noise scale
110 confirms it has written this channel off ($\sigma \approx 8.8$ °C). Binning residuals by load shows that the strong
111 monotone regime bias present under the initial training budget ($+2.1$ to $+5.4$ °C across load bins at
112 the terminal stage) collapses to ± 0.2 °C after retraining with the amended budget: level calibration
113 was a training-sufficiency artefact, not a parameter-count artefact. What remains is structural: the
114 upstream channel’s residual swings ± 8.8 °C across load bins with no monotone signature, pointing at
115 the enthalpy–temperature inversion at the first segment (pressure partition assumption) and the anchor
116 coverage (two spray-outlet temperatures float at initialization), and the first spray-outlet channel
117 retains 44% regime-correlated variance, the signature of unresolved spray transients (Fig. 3). The
118 autoregressive diagnosis generalizes: the process creeps ($0.09\text{--}0.45$ °C per step), so large one-step
119 errors are model jumps, not noise.

120 4 A Discrimination Matrix for World-Model Qualification

121 The tension above is only meaningful if the measurements are reproducible. We froze an evaluation
122 protocol, run end-to-end by a single runner against a versioned data contract, with verdicts computed
123 by paired bootstrap confidence intervals and a 2/3 seed rule.

124 **D-SYN identifiability gate.** The full stack (observer, transition, closure) trains on synthetic data
125 generated by the same architecture and must recover the teacher: 3/3 seeds pass with NLL reductions
126 of 99–136, well above the 30% threshold. Failures on real data therefore reflect structure or data, not
127 an unidentifiable task.

128 **O1 initial-state posterior.** A learned posterior must beat the steady-state anchor by at least 5% with
129 confidence, and not degrade continuity. Side A verdict: the learned posterior is MIXED (two of three
130 seeds degrade significantly, by 30.1% and 31.4% with CIs entirely below zero; one seed improves by
131 13.3%), and the precision-weighted hybrid is REJECTED (no seed reaches a significant improvement:
132 -0.3% , -1.6% , $+3.7\%$, all CIs crossing zero). Under the v0.2 training budget the steady anchor

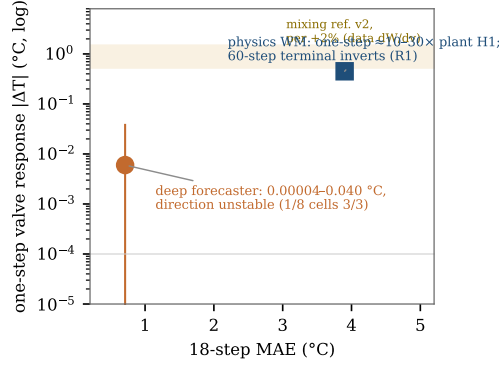


Figure 1: The accuracy–action–fidelity plane (one-step magnitude, log scale). The deep forecaster is accurate but its valve response spans 0.00004–0.040 °C with unstable direction; the physics world model responds on the reference scale at one step (10–30× the plant’s near-zero one-step response) but its 60-step terminal response sign-inverts (R1). Shaded band: data-anchored mixing reference for the secondary valve per +2%.

133 itself improved enough to erase the posterior’s advantage; the learned posterior’s seed instability
134 remains.

135 **T1 transition structure.** Nested arms add closure layers and a latent block against the previous layer.
136 Side A verdicts: the conservative closure is SUPPORTED (2/3 seeds improve val NLL by 5.7% and
137 6.9% with confidence; seed 2 by 2.6%, CI crossing zero). The steam-only placement is REJECTED
138 against it (−6.4%, −2.3%, −3.7%), and the latent block is REJECTED (one seed significantly worse,
139 −1.6%; +0.3%, +3.1% crossing zero). With the v0.2 budget the latent arm converged to the same
140 valley the best seed found before, but not consistently enough to earn its parameters.

141 **B1 boundary forecast.** The learned boundary forecaster is benchmarked against persistence on
142 CRPS. Side A verdict: REJECTED—all three seeds are significantly worse than persistence (CRPS
143 gaps of 1.7–2.0, CIs entirely below zero). Boundary channels are strongly autoregressive; a learned
144 forecaster needs more than a GRU to earn its complexity here.

145 **J1 training scheme.** Joint training of transition and observer against staged training. Side A verdict:
146 SUPPORTED 3/3 (val NLL improvements of 13.5%, 16.4%, 33.3%, all CIs above zero).

147 **R1 action-blind and leakage audit.** The closure must not respond to valve channels (runtime
148 blindness check: 3/3 pass), the boundary whitelist excludes the spray-flow channel, and leakage
149 probes must not read future information (3/3 clean; action-aware probes do not improve, so no
150 leakage is suspected). The direction probe fails: on a 60-step +5% step of the secondary valve only
151 19–34% of windows end negative (the rule as frozen required 100%), with mean terminal deltas of
152 +0.20 to +0.26 °C. Verdict: REJECTED on direction; blind and leakage clean. A protocolized event
153 study on the field data shows the real plant itself ends negative only 68–75% of the time under the
154 same closed-loop confounding, so the frozen rule was stricter than the observable plant; amendment
155 v0.3 recalibrates the direction rule to a mean terminal $\Delta T < 0$ with the bootstrap CI below zero and
156 at least 60% correct windows, reported at both H18 and H60, activated together with the physics
157 repair batch—not retroactively for the current model.

158 Each verdict is computed by the frozen runner, not by hand; the matrix doubles as the artifact we
159 propose as a P5-style honest benchmark for world-model claims [7]. Every number in this section
160 reproduces from the committed artifacts (matrix summary and per-unit reports) through the in-repo
161 replay script. All verdicts above are side A only; the side-B matrix is deferred to a later stage and the
162 single-side scope is carried through the Limitations.

163 5 Discussion

164 The retrained physics line already reaches near-persistence quality on the terminal channel; the
165 remaining cost is concentrated and diagnosable. The upstream channel’s one-step error points at the

660 MW once-through, supercritical dry

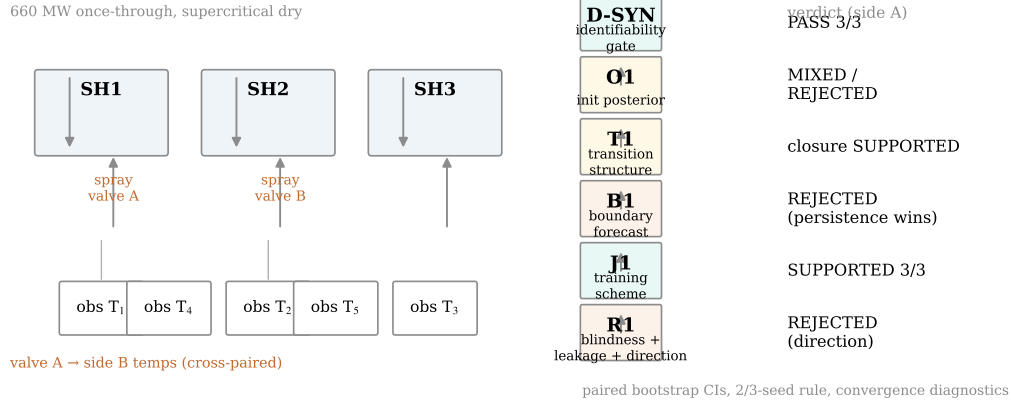


Figure 2: Left: the steam path with five observed temperatures and two cross-paired spray valves. Right: the discrimination matrix with side-A verdicts.

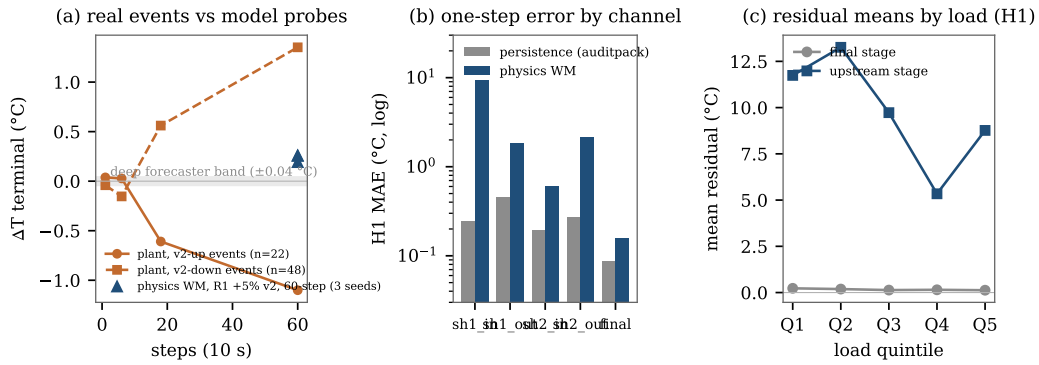


Figure 3: (a) Real secondary-valve events versus model probes: the plant’s response builds over minutes and ends negative on most windows; the physics line’s 60-step terminal response inverts (R1, three seeds); the deep forecaster stays inside ± 0.04 °C. (b) One-step error by channel: the physics line is near persistence at the terminal stage and $38\times$ worse upstream. (c) Residual means by load quintile: the terminal-stage bias is flat after retraining, the upstream stage swings 5–13 °C.

166 initialization and inversion layer, not the dynamics: anchoring all five observations at initialization
167 (two spray-outlet temperatures currently float) and replacing the fixed pressure-partition assumption
168 with a learned or measured one targets most of the 9.4 °C directly. The action-sign failure and the first
169 spray-outlet channel’s residual share one lever: the spray pathway needs transport and evaporation
170 lag in the mixing term, and the rewetting feedback needs a sign-and-magnitude contract (its learned
171 gain over-compensates the direct cooling). Regime-dependent parameter maps, the tempting first
172 remedy, lost their primary evidence once the training budget was corrected—the monotone load bias
173 they were meant to absorb was a training artefact. This ordering illustrates the value of the matrix:
174 one convergence diagnostic and one direction probe re-prioritize the entire optimization agenda.
175 The frozen repair batch (amendment v0.3) executes this ordering: five-point initial-state anchoring
176 with learned pressure-segmented inversion, a spray-to-mixing transport lag state, a hard rewetting
177 sign-magnitude contract, spray-sensitivity priors anchored to the data-fit dW/dv (27.8/70.0 t/h per
178 full travel), with the parameter-map line demoted; the architecture route is a gray-box split by time
179 scale, in which the physics owns the fast dynamics and invariants while learning is restricted to
180 action-blind slow-disturbance estimation and calibration.

181 For deployment we endorse the convergent architecture the reduced-order-model lineage articulates
182 [7, 2, 13]: a learned proposer bounded by a certifiable reduced-order model that approves, bounds,
183 or rejects each proposed action within its certified envelope. Planning with imperfect models is
184 a documented failure mode [10, 15]; the two-tier architecture is the constructive answer. The

185 discrimination matrix is the qualification gate for that proposer. Forecasting remains necessary; it is
186 not sufficient.

187 **Limitations.** Line 1 is teacher-forced, not a recursive foundation forecaster; our claim about recursive
188 forecasters rests on external evidence [14] and should be re-measured with one. Data are from a
189 single unit in dry supercritical operation; two-phase regimes are unobserved. The action-response
190 probe covers valve steps at fixed load; larger interventions and load-coordinated moves deserve their
191 own protocol. All matrix verdicts are single-side (A); the side-B matrix is deferred to a later stage,
192 and no bilateral claim should be read from these results. The quantitative action-response claims
193 use the data-anchored mixing reference; the reference’s regression component is identified under
194 closed-loop operation and carries that caveat.

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